RECYCLE CONVEYOR 2 METAL DETECTOR
2000-F13-PID-PR-10005
SIGNAL

DETAIL 3
HS 13220
XV 13220
FL
F13-DG-467
RECYCLE CRUSHER BYPASS GATE

F13-CR-405
RECYCLE CRUSHER MAINTENANCE CRANE

RECYCLE CONVEYOR 2
2000-F13-PID-PR-10005
PEBBLES
MECH
BYPASS

LI 13221 HH H L LL
LI 13221
LT 13221
LT 13222
LI 13222
LI 13222 HH H L LL

F13-BN-412
RECYCLE CRUSHERS FEED BIN OVERFLOW
MECH

CCTV
F13-CAM-005

F13-BN-411
RECYCLE CRUSHER FEED BIN
O/F
MECH

DE/NDE
TI 13255D HH H
TI 13255E HH H
VI 13254A HH H
VI 13254B HH H

MOTOR RTD
TI 13255A HH H
TI 13255B HH H
TI 13255C HH H
TI 13255F HH H
TI 13255G HH H
TI 13255H HH H

HYDRAULIC UNIT CONTROL PANEL
HS 13279A
HS 13279B
HS 13279C
HS 13279D
HS 13279E
HS 13279F
HS 13279G
HS 13279H
XI 13279I
XI 13279J
XI 13279K

HYDRAULIC UNIT SOLENOID VALVES
HS 13269
HS 13270
HS 13271
HS 13272
HS 13273
HS 13274
HS 13275
HS 13276

F13-HU-403
RECYCLE CRUSHER C HYDRAULIC UNIT

PUMP 1 ISOLATION VALVE
ZAO 13238
PUMP 2 ISOLATION VALVE
ZAO 13239
HYDRAULIC RESERVOIR
LI 13268 H L LL
TI 13267 HH
TRAMP
PI 13264 LL
CLAMP
PI 13263 L LL
BOWL OPEN/CLOSE
PAH 13266
PAH 13265

GATE CLOSED
ZIC 13218
GATE OPENED
ZIO 13218
F13-GA-411
RECYCLE CRUSHER C FEED BIN ISOLATION GATE

F13-GA-412
RECYCLE CRUSHER D FEED BIN ISOLATION GATE
ZIC 13219 GATE CLOSED
ZIO 13219 GATE OPENED

F13-CH-418
RECYCLE CRUSHER C FEEDER CHUTE
F13-FV-401
RECYCLE CRUSHER C FEEDER
VSD
M
SP HS-046 2 OFF
SP HS-046 2 OFF
F13-FV-402
RECYCLE CRUSHER D FEEDER
F13-CH-419
RECYCLE CRUSHER D FEEDER CHUTE

LIC 13223 HH H L
LIC 13227 HH H L
LI 13223
LT 13223
LI 13227
LT 13227

GATE OPEN
HS 13218A
HS 13218C
GATE CLOSE
HS 13218B
HS 13218D

F13-HU-405
RECYCLE CRUSHER ISOLATION GATES HYDRAULIC POWER UNIT

GATE OPEN
HS 13219A
HS 13219C
GATE CLOSE
HS 13219B
HS 13219D

F13-HU-404
RECYCLE CRUSHER D HYDRAULIC UNIT

F13-CH-467
RECYCLE CRUSHER BYPASS CHUTE

F13-CH-461
RECYCLE CRUSHER C FEED CHUTE
LAL 13216
LALL 13216
TAH 13216
TAHH 13216
PDAH 13217
F13-CH-464
RECYCLE CRUSHER D FEED CHUTE

YA 13224
YA 13224
START
YA 13228
YA 13228
START

VSD
M
F13-CC-401
RECYCLE CRUSHER C
F13-CC-402
RECYCLE CRUSHER D

VENDOR PACKAGE 2000-ME001.6-D07-50001

VI 13256
VI 13257
VI 13258
VI 13259
LI 13287 Delete
VI 13316
VI 13317
VI 13318
VI 13319
LI 13287 Delete

DE/NDE
VI 13314A HH H
VI 13314B HH H
TI 13315D HH H
TI 13315E HH H

MOTOR RTD
TI 13315A HH H
TI 13315B HH H
TI 13315C HH H
TI 13315F HH H
TI 13315G HH H
TI 13315H HH H

HYDRAULIC UNIT CONTROL PANEL
HS 13343A
HS 13343B
HS 13343C
HS 13343D
HS 13343E
HS 13343F
HS 13343G
HS 13343H
XI 13343I
XI 13343J
XI 13343K

HYDRAULIC UNIT SOLENOID VALVES
HS 13329
HS 13330
HS 13331
HS 13332
HS 13333
HS 13334
HS 13335
HS 13336

BOWL OPEN/CLOSE
PAH 13325
PAH 13326
CLAMP
PI 13323 LL
TRAMP
PI 13324 LL
HYDRAULIC RESERVOIR
TI 13327 HH
LI 13328 H L LL
PUMP 1 ISOLATION VALVE
ZAO 13298
PUMP 2 ISOLATION VALVE
ZAO 13299

LUBE OIL SUPPLY
FI 13244
TI 13242 HH H L LL
LUBE OIL TANK
LI 13241 L LL
LUBE OIL RETURN
TI 13243 HH H
TDI 13247
LUBE OIL SUPPLY
TI 13247 L
LUBE OIL FILTER
PI 13237
PDI 13237 H
PI 13236

F13-LU-403
RECYCLE CRUSHER C LUBRICATION UNIT

25VB03S3 F13-10398
25VB03S3 F13-10399
25VB03S3 F13-10400
25VB03S3 F13-10401
25VB03S3 F13-10402
13-0914-LO-SS10-50
13-1274-LO-SS10-50
13-0916-LO-SS1-100
13-0901-LO-SS3-25
SP MS-003
HPU FILTER SWITCH PDAH 13260
PUMP DISCHARGE PI 13261
HPU FILTER PDI 13261 H
SYSTEM PRESSURE PI 13262
10°
25VB03S3 F13-10396

F13-FN-457
RECYCLE CRUSHER C AIR COOLER 1
F13-FN-458
RECYCLE CRUSHER C AIR COOLER 2

13-0915-LO-SS10-50
PRV 13235
13-1435-PSW-SS3-25
PI 13188
25VB03S1 F13-10201
FI 13187 L LL
FIT 13187
NOTE 1
SET PRESSURE @ 100kPag
PCV 13186
DETAIL 11
HS 13344
SV 13344
S
FC
50x25
13-1166-FD-PE8-50
13-1176-FD-PE8-50
90x50
TO GRADE
ZA 13252A
ZA 13252B
BOWL POSITION
PI 13233
LUBE OIL SUPPLY
TI 13232 HH H
SI 13253 L LL
COUNTER-SHAFT
F13-CH-462
RECYCLE CRUSHER C DISCHARGE CHUTE

SI 13313 L LL
TI 13282 HH H
COUNTER-SHAFT
F13-CH-465
RECYCLE CRUSHER D DISCHARGE CHUTE
PI 13283
LUBE OIL SUPPLY
ZA 13312B
ZA 13312A
BOWL POSITION

PDAH 13320
HPU FILTER SWITCH
PI 13321
PUMP DISCHARGE
PDI 13321 H
HPU FILTER
PI 13322
SYSTEM PRESSURE
10°
SP MS-003
25VB03S3 F13-10403
25VB03S3 F13-10397
13-0902-LO-SS3-25
13-0919-LO-SS1-100
13-1275-LO-SS10-50
13-0917-LO-SS10-50
25VB03S3 F13-10404
25VB03S3 F13-10406
25VB03S3 F13-10405
25VB03S3 F13-10407
13-0918-LO-SS10-50
13-1182-FD-PE8-50
13-1223-FD-PE8-50
90x50
TO GRADE
PRV 13285
25VB03S1 F13-10206
PI 13295

F13-LU-404
RECYCLE CRUSHER D LUBRICATION UNIT
F13-FN-459
RECYCLE CRUSHER D AIR COOLER 1
F13-FN-460
RECYCLE CRUSHER D AIR COOLER 2
VENDOR PACKAGE 2000-ME001.6-D07-50001

LUBE OIL SUPPLY
FI 13304
TI 13302 HH H L LL
LUBE OIL TANK
LI 13301 L LL
LUBE OIL RETURN
TI 13303 HH H
TDI 13307
LUBE OIL SUPPLY
TI 13307 L
LUBE OIL FILTER
PI 13297
PDI 13296 H
PI 13296

RAW WATER PUMPS
2000-F24-PID-PR-10018
RAW WATER
13-2046-BW-SS3-50
50VB03S1 F13-10373
50VDE1S1 F13-10375
25VB03S1 F13-10374
TO GRADE
DETAIL 11
HS 13185
SV 13185
FO
DETAIL 11
HS 13345
SV 13345
S
FC
13-2047-BW-SS3-25
SP SP-005
SP SP-005
BRA 13354
BRS 13354
BDA 13353A
BDS 13353A
BDA 13353B
BDS 13353B
PWA 13350A
PWS 13350A
PWA 13350B
PWS 13350B
SSL 13351
SAL 13351
F13-CV-406
RECYCLE CRUSHER DISCHARGE CONVEYOR

START
YA 13355
YA 13355
VSD
M
F13-CH-463
RECYCLE CRUSHER DISCHARGE CONVEYOR HEAD CHUTE
LSH 13356
LAH 13356
SAL 13352
ENCODER
ZE 13352
VE 13357A
VE 13357B
VI 13357A HH H
VT 13357A
VT 13357B
VI 13357B HH H
TI 13357A HH H
TI 13357B HH H

FI 13294 LL
FIT 13294
NOTE 1
SET PRESSURE @ 100kPag
PCV 13293
DETAIL 11
HS 13292
SV 13292
FO
SAG MILL FEED CONVEYOR
2000-F13-PID-PR-10001
CRUSHED PEBBLES
25VB03S1 F13-10205
TO GRADE
25VB03S1 F13-10204

SCHEME WATER DISTRIBUTION
2000-F24-PID-PR-10006
SCHEME WATER
13-1434-PSW-SS3-50
13-0770-PSW-SS3-25

NOTES:

1. FLOWMETER HAS DN25 BSP FEMALE THREAD CONNECTIONS.

DRG No.	REFERENCES
2000-F0-PID-PR-10007	LEGEND SHEET 7 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F0-PID-PR-10006	LEGEND SHEET 6 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F0-PID-PR-10005	LEGEND SHEET 5 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F0-PID-PR-10004	LEGEND SHEET 4 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F0-PID-PR-10003	LEGEND SHEET 3 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F0-PID-PR-10002	LEGEND SHEET 2 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F0-PID-PR-10001	LEGEND SHEET 1 - PIPING AND INSTRUMENTATION DIAGRAM

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
1	12SEP25	GENERAL REVISIONS ADDED	[redacted]				
0	28OCT24	APPROVED FOR CONSTRUCTION					

NORTHERN STAR RESOURCES LIMITED

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PRIMERO PROJECT NUMBER: 23517

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: MILLING & CLASSIFICATION
PEBBLE CRUSHING - SHEET 2 - STAGE 1
PIPING AND INSTRUMENTATION DIAGRAM

SHEET	SCALE	DRG No.	REVISION
A1	NTS	2000-F13-PID-PR-10006	1

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\PRP\PIDS\ISSUED\12.09.25 - SIGNING\NATIVES\2000-F13-PID-PR-10006_1.DWG 10SEP25 10:04AM ANNE.HOYNE